

05_Backend_Schema

Agentic Regulatory Governance & Unified Simulation

Team Rocket

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1. Schema Overview (PostgreSQL)

All tables include id UUID PK, org_id UUID FK, created_at, updated_at unless noted. Multi-tenant isolation via org_id on every table.

2. Core Tables

2.1 organizations

Column	Type	Notes
id	UUID PK	
name	text	
industry	text	
created_at	timestamptz	

2.2 departments

Column	Type	Notes
id	UUID PK	
org_id	UUID FK → organizations	
name	text	e.g., “Risk Management”, “IT Security”

2.3 users

Column	Type	Notes
id	UUID PK	
org_id	UUID FK	
email	text unique	
password_hash	text	
role	enum	admin / compliance_officer / department_head / auditor / viewer
department_id	UUID FK, nullable	for department_head role

2.4 circulars

Column	Type	Notes
id	UUID PK	
org_id	UUID FK	

title	text	
source_file_url	text	
effective_date	date	
status	enum	uploaded / obligations_extracted / mapped / stress_tested / remediated
raw_text	text	full extracted text
uploaded_by	UUID FK → users	

2.5 obligations

Column	Type	Notes
id	UUID PK	
circular_id	UUID FK → circulars	
description	text	
deadline	date, nullable	
applicability	text	e.g., “all listed entities”
source_ref	text	section/clause reference in circular
status	enum	ai_extracted / confirmed / edited / manually_added
confirmed_by	UUID FK → users, nullable	

2.6 policies

Column	Type	Notes
id	UUID PK	
org_id	UUID FK	
title	text	
document_url	text	
department_id	UUID FK, nullable	
last_updated	date	used for “outdated procedure” check
document_type	enum	policy / sop / workflow_definition

2.7 obligation_policy_mappings

Column	Type	Notes
id	UUID PK	
obligation_id	UUID FK → obligations	
policy_id	UUID FK → policies, nullable	null = unmapped
department_id	UUID FK → departments	
confidence	numeric(3,2)	0.00–1.00, AI mapping confidence
mapping_source	enum	ai / manual
confirmed_by	UUID FK → users, nullable	

2.8 findings

Column	Type	Notes
id	UUID PK	
circular_id	UUID FK	
obligation_id	UUID FK → obligations	
type	enum	unimplemented / outdated_procedure / policy_conflict / workflow_gap / missing_evidence
severity	enum	high / medium / low
description	text	
status	enum	open / in_progress / resolved
detected_at	timestampz	

2.9 evidence

Column	Type	Notes
id	UUID PK	
org_id	UUID FK	
finding_id	UUID FK → findings, nullable	can be uploaded pre-emptively
obligation_id	UUID FK → obligations, nullable	
document_url	text	
document_type	enum	policy_document / training_record / compliance_report / approval / audit_document
status	enum	present / stale / missing
uploaded_by	UUID FK → users	
verified_at	timestampz, nullable	

2.10 regulatory_replays

Column	Type	Notes
id	UUID PK	
finding_id	UUID FK → findings	
chain_json	jsonb	ordered array of {entity_type, entity_id, label} nodes
explanation	text	ARGUS Advisor generated narrative
generated_at	timestampz	

2.11 readiness_scores

Column	Type	Notes
id	UUID PK	
org_id	UUID FK	
circular_id	UUID FK, nullable	null = org-wide aggregate score
overall_score	numeric(5,2)	0-100
policy_alignment	numeric(5,2)	
control_coverage	numeric(5,2)	

evidence_completeness	numeric(5,2)	
workflow_readiness	numeric(5,2)	
employee_readiness	numeric(5,2)	
computed_at	timestamptz	append-only, enables trend charting

2.12 action_items

Column	Type	Notes
id	UUID PK	
finding_id	UUID FK → findings	
owner_department_id	UUID FK → departments	
priority	enum	high / medium / low
suggested_deadline	date	
status	enum	not_started / in_progress / done
updated_by	UUID FK → users, nullable	

2.13 reports

Column	Type	Notes
id	UUID PK	
org_id	UUID FK	
circular_id	UUID FK, nullable	
file_url	text	generated PDF
generated_at	timestamptz	
generated_by	UUID FK → users	

3. Vector Store (ChromaDB / FAISS)

Separate from Postgres — holds embeddings for semantic retrieval, keyed by org_id:

Collection	Chunk source	Metadata stored
circular_chunks	circulares.raw_text (chunked)	circular_id, section_ref
policy_chunks	policies.document_url (parsed, chunked)	policy_id, department_id, last_updated
evidence_chunks	evidence.document_url (parsed, chunked)	evidence_id, document_type

Used by: Process Mapping Agent (obligation ↔ policy retrieval), Evidence Verification Agent (evidence ↔ obligation retrieval), ARGUS Advisor (RAG Q&A grounding).

4. Key Relationships (ER Summary)

```
organizations 1-* users
organizations 1-* departments
organizations 1-* circulars
circulars 1-* obligations
obligations *-* policies    (via obligation_policy_mappings)
obligations 1-* findings
```

```
findings 1-1 regulatory_replays
findings 1-* evidence
findings 1-* action_items
organizations 1-* readiness_scores (append-only time series)
organizations 1-* reports
```

5. Indexing & Constraints Notes

- obligations.circular_id, findings.circular_id, readiness_scores.org_id + computed_at should be indexed — dashboard queries filter/sort on these heavily.
- Unique constraint on (obligation_id, policy_id) in obligation_policy_mappings to prevent duplicate mappings.
- readiness_scores is append-only (never updated in place) to preserve trend history for the dashboard trend line.
- All file references (source_file_url, document_url, file_url) point to object storage (e.g., S3-compatible bucket), not stored as blobs in Postgres.